IT2021 AIML Project – 2026 Jan

Topic Approval Form

1. Project ID: AI-02-G03

2. Project Title: **Smart Clinic:** AI-Powered Private Medical Centre Management System

3. Campus: Malabe

4. WD/WE: WD

5. Group Information:
   Stream: AI

	Registration Number	Student Name	Phone Number	Signature
01	IT24201016	Wijewardhana E.A.N	0714692373	
02	IT24104107	Dilsha W.K.C	0705680562	
03	IT24103228	Fonseka N.H.D.K	0712469215	
04	IT24103073	Vithusa K	0759705656	
05	IT24101175	Premarathne H M H S	0717784225	
06	IT24102346	Dewanarayana S D W	0711588975	

6. Client Information / Project Justification
   A. Project with a Client:
      Client Name/ Organization: Family Medical Centre (Dr.Dilini Indunil)
      Contact Person & Designation: N.H.D.I Fonseka/Owner
      Email: indunilfonseka97@gmail.com
   B. Projects without a Client: (Briefly justify the selection of the project title.)

7. A brief description of the problem.

Small medical clinics in Sri Lanka currently face operational bottlenecks due to manual, paper-based management of patient records, appointments, and prescriptions. These inefficiencies result in scheduling conflicts and, more critically, potential patient safety risks such as overlooked drug–drug interactions and delayed symptom triage. The proposed solution, Smart Clinic, replaces these manual workflows with an intelligent digital platform that incorporates a symptom-based disease predictor and an automated drug interaction checker. By integrating machine learning algorithms to analyze patient symptoms and provide clinical decision support, the system significantly enhances diagnostic accuracy and patient safety. This transition from a passive record-keeping system to an active, intelligent assistant directly improves clinic productivity and the overall quality of healthcare delivery.

8. Main Features of the Proposed System.

List software features. Software features should be functional and meaningful parts of the system, not just simple UI pages. Where possible, they should connect to or use the AI/ML feature.

	Registration Number (Same order as above)	**Name of Feature/s**	**Brief description of feature(s) in point form**
01	IT24201016	Medical Records & Consultation Management	• Manage electronic medical records (EMR) and historical consultation notes. Implement CRUD operations for clinical vitals such as weight, blood pressure, and temperature with basic trend visualization. • Securely link medical records with appointments and patient profiles.
02	IT24104107	Appointment Scheduling & Billing Management	• Book, update, and cancel appointments with calendar-based validation to avoid conflicts. • Record consultation fees generate digital receipts, and track payments and outstanding dues. • Send automated appointment reminders to patients via notifications.
03	IT24103228	Patient and Staff Management	• Register, update, search, and securely view patient profiles including demographics and medical history. • Manage staff and doctor profiles, roles, and availability. • Implement unified role-based access control and privacy management for patients and staff.
04	IT24103073	Prescription Management & Drug Safety	• Create, update, and search digital prescriptions in a structured format. • Integrate automated drug interaction checking to identify unsafe medication combinations during prescribing. • Generate professional PDF prescriptions with safety warnings and clinical disclaimers.

05	IT24101175	Patient Awareness Alert System	• Create health alerts, events, and announcements for patients. • Target patients based on criteria such as age, condition, or visit history. • Send personalized email and SMS notifications.
06	IT24102346	Patient Review & Feedback System	• Collect post-visit patient feedback including ratings and comments. • Display aggregated feedback trends on the clinic dashboard. • Support moderation, replies, and reporting to improve service quality.

9. AI/ML Feature of the Proposed System.

Feature Name: **AI-Assisted Symptom Analysis and Drug Interaction Detection**

	Registration Number (Same order as above)	**Expected individual contribution to the AI/ML feature.**
01	IT24201016	**Machine Learning Model Development:** Designing, training, and evaluating the symptom-based disease prediction model, selecting suitable classification algorithms, and validating model accuracy and performance.
02	IT24104107	**Drug Interaction Data Preparation & API Integration:** Sourcing, standardizing, and organizing public drug–drug interaction datasets. Mapping medication names to standardized formats and integrating a third-party Drug Interaction Checker API to provide structured interaction data for use in the prescription workflow. Supports the system by delivering clean, actionable data for severity classification and real-time safety alerts.
03	IT24103228	**Data Engineering for Symptom Analysis:** Collecting public medical symptom–disease datasets (e.g., Kaggle), performing data cleaning, handling missing or inconsistent values, encoding symptoms, and converting datasets into CSV/ARFF formats suitable for machine learning model training.
04	IT24103073	**AI-Enhanced Prescription Safety & Symptom Analyzer Integration:** Designing, training, and evaluating a machine learning classification model to predict the severity level of drug–drug interactions (e.g., mild, moderate, severe) using standardized medication data. Integrating the model's predictions into the digital prescription module to generate real-time safety warnings.

		Designing the symptom input interface for the Symptom Analyzer, integrating model predictions with probability scores, and implementing real-time display with clinical disclaimers to support doctors during consultations.
05	IT24101175	**Patient Awareness & AI-Driven Notification Logic:** Supporting AI-assisted patient communication by targeting patient groups based on medical or visit criteria and integrating alert mechanisms to deliver personalized health notifications related to clinic events and reminders.
06	IT24102346	**Review Analysis & System Validation:** Implementing basic sentiment categorization (keyword or rule-based) for patient feedback to identify service trends, adding disclaimers for AI-generated insights, and conducting end-to-end testing and validation of AI features.

10. Approval by the Evaluator

Name: Arkini Thayanadhone / Sachintha Wathasingh

Date: 6/2/26